

CapstoneProjectCharter

Project Charter

1. Project Overview

Project Name: D&D LLM campaign

Project Lead / Initiator: Abdulfetah Adem, Mizpah Parrilla, Spencer Kunz

Date: Feb 16, 2026

Executive Summary

The D&D LLM Campaign project is a multi-agent AI storytelling system where each Dungeons & Dragons character is powered by a Large Language Model (LLM). The system simulates a full campaign experience including narrative progression, autonomous character decision-making, rule enforcement, and persistent world memory.

The project solves the problem of limited availability of human Dungeon Masters and inconsistent AI storytelling by creating a structured, memory-aware multi-agent system.

The intended users (The platform target) include tabletop RPG players, AI enthusiasts, and researchers interested in multi-agent LLM system and Generative AI

Primary Goal

By the end of Spring Quarter, deliver a fully functional web-based prototype that supports:

- Multi-agent gameplay (DM + 3–4 characters)
- Persistent shared and private memory
- Rule enforcement and state tracking
- Dynamic storytelling and interaction
- Session recap and logging

2. The "Consultancy" Requests

This section defines the work required from each specific department. This acts as the "Scope of Work" for the student teams.

A. Business Department Needs (International Business)

We need the Business students to investigate and provide guidance on the following:

1. **Target Market:** (e.g., Who is our ideal customer? What is the demographic?)

Primary: Tabletop RPG players (ages 18–40)

Secondary: AI enthusiasts and developers

Tertiary: Researchers studying multi-agent AI system

2. **Competitive Analysis:** *(e.g., Who else is doing this? Why will we fail or succeed against them?)*

- Competitors:
 - AI Dungeon (text-based storytelling)
 - ChatGPT-based RPG experiments
- Competitive Advantage:
 - Multi-agent architecture (no single AI)
 - Persistent structured memory
 - Rule-based validation layer
- Risks:
- Existing platforms have larger user bases
- Rapid competition in AI tools

3. **Business Model:** *(e.g., How does this make money? Subscription? Ads? Free service?)*

- Freemium model:
 - Free tier: limited sessions
 - Paid tier: extended campaigns + premium features
- Optional:
 - Subscription (\$5–\$15/month)
 - API access for developers
 - Premium character packs

4. **Specific Deliverables Requested:**

- a. ☒ Market Research Report
- b. ☒ SWOT Analysis
- c. ☒ Marketing/Go-To-Market Strategy
- d. ☐ Other: Monetization Strategy Documentation

B. Technical Needs (App Dev & Computer Science)

We need the technical students to build or analyze the following:

1. **Platform:**

Web-based application using:

- frontend: Streamlit or React
- Backend: FastAPI
- LLM: OpenAI / LLaMA
- Memory: FAISS or Chroma

2. **Key Features (MVP):**

- a. Multi-Agent System (DM + 3-4 Character LLM's)
- b. Dice roll simulation
- c. Character tracking sheet
- d. HP, inventory, abilities
- e. Persistent shared memory
- f. Turn-based interaction system
- g. Session logging and recap generation

3. **Data Requirements:**

- store campaign logs
- store character sheets
- store world state (locations, quests, etc)
- API key management
- Memory embeddings (vector database)

4. **Specific Deliverables Requested:**

- a. ☐ Functional Prototype / MVP
- b. ☐ Database Schema & API Documentation
- c. ☐ Code Repository (GitHub/GitLab)
- d. ☐ Deployment Plan

C. Graphic Design Needs

Consider any graphic design needs that might be required:

1. **Design scope:**

- Brand identity
- UI/UX design
- Game interface layout
- Visual storytelling elements

2. **Key Components (MVP):**

- a. [Component 1 – e.g., Logo and brand mark]
- b. [Component 2 – e.g., Color palette and typography system]
- c. [Component 3 – e.g., UI mockups or wireframes for core screens]

3. **Design Requirements:**(e.g, Designs should align with brand values, be accessible, and follow usability best practices such as contrast ratios and responsive layouts.)

- Accessible (contrast, readability)
- Responsive design (desktop-first)
- Clean and immersive UI
- Consistent visual hierarchy

4. **Target Audience & Use Context:** (e.g., General consumers, students, internal staff; mobile-first vs desktop-first usage)

- Desktop users (primary)
- Students, gamers, developers
- Casual and advanced users

5. Specific Deliverables Requested:

- a. [x] Brand Style Guide (logo usage, colors, fonts)
- b. [x] High-fidelity Mockups (Figma, Adobe XD, etc.)
- c. [x] UI Assets (icons, buttons, illustrations)
- d. [x] Marketing or Presentation Graphics (optional)
- e. [x] Exported Assets for Developer Handoff

3. **Operational Details**

A. Existing Assets

(Are there existing logos, color palettes, previous codebases, or raw data files? Link to them in the OneDrive folders here.)

1. [Link to design folder]
2. [Link to previous code]

Currently none (project starts from scratch)

Planned:

- GitHub repository (source code)

- Figma design files
- Dataset logs (generated gameplay data)

B. Success Metrics

(How will we know we succeeded at the end of the quarter?)

1. *Business Success:*
 - a. 3 monetization pathways
 - b. validate target market (research, surveys etc.)
 - c. have competitor positioning
2. *Technical Success:* [e.g., Successful end-to-end test of a user purchase.]
 - a. fully functional multi-AI agent system run
 - b. persistent memory recall across turns
 - c. character sheet updates automatically
 - d. successful save/load functionality
 - e. System response time < 3 seconds

4. Team Roster (Consultants)

Please list the students assigned to this project folder.

Department	Student Name(s)	Contact Email
Project Lead	Abdulfetah Adem	abdulfetah.adem@seattlecolleges.edu
Business Team	[Names]	[Email]
App Dev Team	[Names]	[Email]
Comp Sci Team	Abdulfetah Adem, Mizpah Parrilla, Spencer Kunz	mizpah.parrilla@seattlecolleges.edu , spencer.w.kunz@gmail.com
Design Team	[name]	[email]

5. Communication Plan (Consultants)

Please give details on how communication will occur both internally with the consultancy team and with the client.

Stakeholder: People who have a vested interest in the project.

Ex: CS Development Team, Business Team, Instructors

Format: Preferred format for communication.

Ex: Slack, Email, Microsoft Teams, or Status Report.

Frequency: How often the communication happens.

Ex: Once a week, Every few weeks

Owner: The Person responsible for making sure the Communication goes out.

Internal Communication Plan

Stakeholder	Format	Frequency	Owner
Dev Team	Slack / Discord		Tech Lead
Business Team	Teams / Email		Business Lead
Design Team	Figma / Slack		Design Lead

Communication with Client/Instructor

Stakeholder	Format	Frequency	Owner

6. Timeline (Consultants)

Provide a rough idea of a timeline starting from mid-Winter quarter until the end of Spring quarter. Include planned tasks to be completed.

March 1 – March 14

Goals:

- Finalize system architecture diagram
- Define JSON schema for DM + character outputs
- Choose tech stack (OpenAI/LLaMA, FAISS/Chroma, Streamlit/FastAPI)
- Create GitHub repo + folder structure
- Build single-agent DM prototype (text-based)
- Define character stat structure (HP, inventory, abilities)

Milestone:

Single-agent playable prototype (no memory yet)

March 15 – March 28

Goals:

- Implement multi-agent orchestration (DM + 3–4 characters)
- Build turn-based interaction loop
- Implement structured JSON parsing + validation
- Add dice roll simulator
- Implement character sheet tracking

March 29 – April 18

Goals:

- Integrate vector database (FAISS/Chroma)
- Implement shared campaign memory
- Implement private agent memory
- Add memory retrieval to prompts
- Implement save/load campaign state
- Add session logging

Milestone:

System recalls past events across sessions

April 19 – May 9

Goals:

- Add rule enforcement validation layer
- Add contradiction detection (basic checks)
- Improve prompt consistency
- Implement session recap generator
- Build UI panels (Character Sheet, Memory, Quest Log)
- Begin business analysis deliverables

Milestone:

Stable playable prototype with persistent memory

May 10 – May 24

Goals:

- Final UI polish (Streamlit layout improvements)
- Improve agent personality differentiation
- Add hidden goals (optional stretch)
- Performance optimization
- User testing (run multiple campaign simulations)
- Finalize business & design documentation

Milestone:

Showcase-ready build

May 25 – June 10

Goals:

- Bug fixes
- Documentation (README + Architecture Explanation)
- Create demo script
- Record backup demo video
- Prepare final presentation slides
- Prepare technical report

Final Deliverable Due:

Fully functional DND LLM Campaign system with live demo capability

